

CPU Rolling-Contact Support

Complete rigid-wheel rolling contact for differential-drive and four-steering robots, implemented through the native Tasks and TVM CPU solver paths and validated in headless MuJoCo.

● Complete · verified · committed

102/102

complete repository
CTests passed

250

physical CPU MuJoCo
processes

0

GPU runtime
dependencies or
deadline misses

0.201 ms

worst representative 10k-
cycle p99

OUTCOME

Executive summary

The rolling-contact roadmap is complete. `mc_rtc` now provides a backend-neutral rolling-contact geometry kernel, native Tasks and TVM constraint/dynamics implementations, two installable wheeled robots, a production sample controller, and a reproducible CPU-only validation suite. The implementation runs without discovering, requiring, or linking a GPU runtime.

Release decision: pass. All roadmap stages are closed; the exact committed snapshot builds and passes the rolling integration suite, the complete regression suite, sanitizer checks, deterministic physical simulation, long-duration timing, allocation, and dynamic-linkage gates.

2

Native CPU backends: Tasks
QLD and TVM weighted least
squares

2

Robot families: differential drive
and four independently
steering/driving wheels

25

Physical cases per backend,
including ramps, disturbances,
low friction, and mode changes

CPU ROLLING-CONTACT IMPLEMENTATION

e21ed9a997ed73bafbb345a82f92bcdfb1c28585
feat: add CPU rolling-contact support

UBUNTU 24.04 BUILD SUPPORT PARENT

fc7df14a96f1273fb1aaff8e1a84ef514455d29c
fix: support Ubuntu 24.04 builds

DELIVERED SCOPE

Implementation inventory

Geometry and contact model

- Finite wheel-line contact geometry and right-handed rolling frame $[\mathbf{t}, \mathbf{l}, \mathbf{n}]$.
- Velocity/acceleration no-slip rows, bias, Baumgarte stabilization, and finite-input validation.
- Two force endpoints with four-ray friction cones, wrench diagnostics, and wheel-torque evaluation.
- Zero heap allocations across 10,000 hot-loop updates after warm-up.

Solver integration

- Native hard and soft longitudinal rolling for Tasks and TVM.
- Persistent contact identities across numerical updates and mode transitions.
- Actuated-only TVM torque variable; floating-base efforts remain zero.
- Deferred TVM graph-object retirement prevents cached-solver lifecycle use-after-free.

Robots and controller

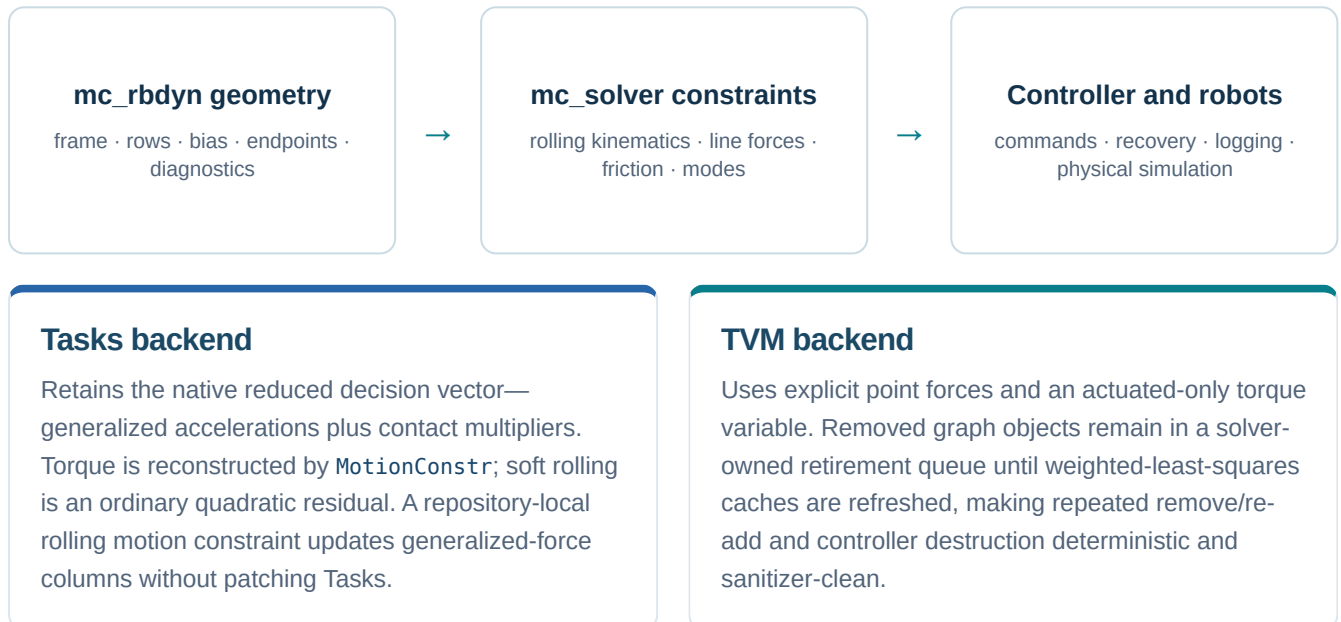
- Differential and four-steering URDF, RSDF, aliases, gains, and MuJoCo descriptions.
- Forward/reverse, turns, circles, sinusoid, crab, Ackermann, pure-yaw, and steering-rate commands.
- Fixed, rolling, sliding, and detached modes with deterministic transition coverage.
- Infeasibility detection, fallback/recovery, logging, datastore, and GUI diagnostics.

Handoff and evidence

- Geometry, robot, solver, controller, invalid-input, parity, lifecycle, and allocation tests.
- Headless pinned CPU MuJoCo runner and deterministic scenario matrix.
- CPU dynamic-link audit rejecting CUDA, cuBLAS, cuSolver, Torch, HIP, and OpenCL.
- English/Japanese tutorials, JSON schemas, QP report, architecture record, and retained JSON evidence.

ARCHITECTURE

One contact model, two CPU solver paths



Verification matrix

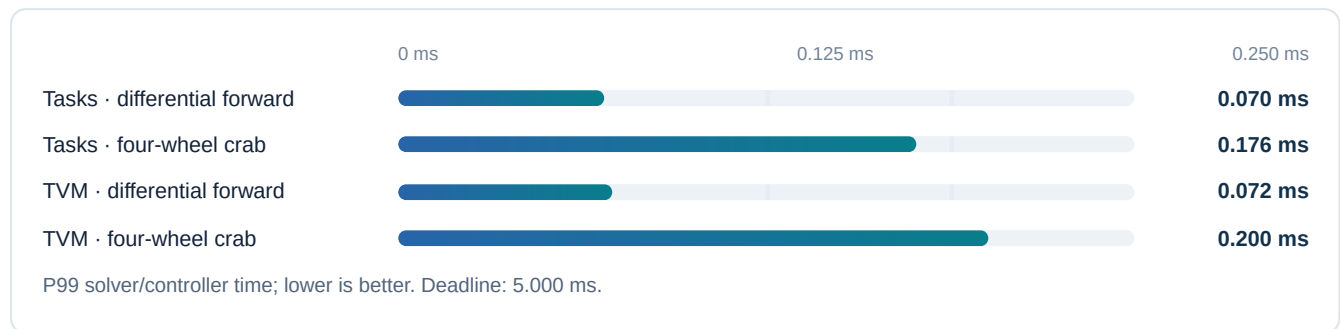
GATE	RESULT	DIRECT EVIDENCE
Exact committed snapshot	PASS	Isolated configure/build; 13/13 rolling CTests
Complete regression	PASS	102/102 CTests in 34.93 seconds (GUI sockets disabled)
AddressSanitizer	PASS	4/4 targeted families; no sanitizer failure
Lifecycle stress	PASS	50 controller lifecycle and 20 complete solver repetitions
Allocation hot loop	PASS	10,000 updates; zero allocations after warm-up
CPU dynamic linkage	PASS	5 targets; zero GPU runtimes; zero unresolved libraries
Physical MuJoCo matrix	PASS	250 processes; zero failures or missed deadlines
10,000-cycle timing	PASS	4 representative cases; every p99 below 0.201 ms
Formats and documentation	PASS	JSON, YAML, XML, Python, shell, Doxygen, schemas, EN/JP tutorials

Backend numerical parity

QUANTITY	MEASURED ERROR	ACCEPTANCE
Rolling coefficient matrix	0	1e-10 absolute
Acceleration bias	0	1e-10 absolute
Generalized acceleration	3.5304e-7	1e-6 scaled
Actuated torque	6.6089e-16	1e-7 scaled
Resultant contact wrench	4.2365e-6	1.1868e-5 scaled
TVM soft-rolling residual	4.9828e-10	residual policy

Long-duration CPU timing

Four production cases ran 100 warm-up cycles followed by 10,000 measured cycles at a 5 ms controller period. No case missed a deadline. The chart uses a 0.25 ms p99 scale; the actual deadline is twenty times that scale.



BACKEND / CASE	MEDIAN MS	P95 MS	P99 MS	MAX MS	RTF	RSS KIB
Tasks / differential forward	0.044740	0.064231	0.070081	0.226873	86.1634	50,648
Tasks / four-wheel crab	0.118021	0.151712	0.175613	0.238283	35.5080	51,648
TVM / differential forward	0.047081	0.058371	0.072461	0.145982	83.3322	52,148
TVM / four-wheel crab	0.136831	0.175352	0.200392	0.357224	31.0695	52,672

CPU MuJoCo validation

The pinned headless runner exercised 25 named scenarios on both backends. Each backend/scenario pair ran five deterministic repetitions: 250 processes, each with 100 warm-up and 1,000 control cycles.

250

independent physical processes

0

controller or validation failures

0

missed 5 ms deadlines

20.53×

minimum real-time factor

Differential scenarios

Hold, forward, reverse, left/right turn, left/right circle, sinusoid, ramp up/down, low friction, lateral impulse, contact loss, and the complete fixed/rolling/sliding/detached mode cycle.

Four-steering scenarios

Forward, reverse, crab, left/right Ackermann, pure yaw, nonzero steering rate, ramp crab, lateral impulse, low friction, and the complete contact-mode cycle.

PHYSICAL-SUITE AGGREGATE	RESULT
Worst observed p99 controller time	0.346865 ms
Minimum real-time factor	20.528409×
Maximum resident memory	52,312 KiB
Backend/case pairs accepted	50 / 50
Repetitions accepted	250 / 250



CPU-only contract proven

All executable gates ran with `CUDA_VISIBLE_DEVICES=-1`. Five production/integration targets resolved every dynamic library and had empty CUDA, cuBLAS, cuSolver, Torch, HIP, and OpenCL dependency lists. There is no rolling-contact GPU build option or fallback path.

Recorded CPU configuration

Operating system	Ubuntu 24.04, x86_64
Processor	AMD Ryzen 9 9950X; 16 physical cores / 32 logical CPUs; SMT2; boost enabled
Toolchain	GCC 13.3.0, CMake 3.28.3, C++17, RelWithDebInfo
Solvers	Tasks 1.8.4 QLD CPU path; TVM 0.9.5 default weighted least-squares CPU path
Simulation	MuJoCo 3.3.6; mc_mujoco revision 9397c73f with tracked user-destination patch
Timing setup	Single controller process; single-threaded solve/controller path; 5 ms period; no asynchronous device work

Core release checks

```
cmake -S . -B build -DCMAKE_BUILD_TYPE=RelWithDebInfo -DBUILD_TESTING=ON
cmake --build build -j2

CUDA_VISIBLE_DEVICES=-1 MC_RTC_DISABLE_CONVEX_GENERATION_PATCH=ON \
  ctest --test-dir build -R 'RollingContact|testRollingContact' \
  --output-on-failure -j1

MC_RTC_CONTROLLER_CONFIG=/tmp/mc_rtc-no-gui.yaml CUDA_VISIBLE_DEVICES=-1 MC_RTC_DISABLE_CONVEX_GENERATION_PATCH=ON \
  ctest --test-dir build --output-on-failure -j1

ctest --test-dir build -R '^RollingContactCPULinkage$' --output-on-failure
```

Physical suite

```
CUDA_VISIBLE_DEVICES=-1 rolling-contact-report/scripts/build-cpu-mujoco-runner.sh
CUDA_VISIBLE_DEVICES=-1 python3 rolling-contact-report/scripts/run-mujoco-suite.py \
  --runner /tmp/rolling-contact-cpu-mujoco/runner-build/rolling_contact_mujoco_runner \
  --build build --artifact-dir /tmp/rolling-contact-mujoco-results --repetitions 5
```

Visible four-steer model and keyboard teleoperation

The four-steering examples now select the one-argument `RollingContactRangerMiniV3` alias. The tracked MuJoCo model is self-contained (no unshipped V2 meshes), uses a 0.494 m wheelbase, 0.364 m track, 0.125 m wheel radius, and has four independently steered and driven wheels.

MuJoCo contact hygiene

The wheels sit inside the visible body envelope. Explicit chassis/wheel collision exclusions prevent artificial self-contact impulses before the first QP solve while retaining wheel/terrain contacts.

RoboticsUtils keyboard

A real terminal starts `KeyboardCapture` for the `keyboard` scenario. W/S drive forward/reverse, A/D crab left/right, Q/E yaw left/right, `space` clears all latched axes, and X stops capture and zeros the controller references. Each axis is latched until its opposite key or `space` is pressed; the Tasks backend sends chassis and wheel-rate feed-forward to the position/orientation/posture tasks on the next control cycle. The controller synchronizes MuJoCo encoder and `FloatingBase` measurements into its MBC, then emits measured-state-relative wheel `q/alpha` references, matching the signals consumed by `mc_mujoco`. Ctrl-C or closing the MuJoCo window terminates the simulator.

```
source /opt/ros/jazzy/setup.bash
export CUDA_VISIBLE_DEVICES=-1 MC_RTC_DISABLE_CONVEX_GENERATION_PATCH=ON
export LD_LIBRARY_PATH="$PWD/build/src:$PWD/build/plugins/ROS:$PWD/build/deps/tasks-system-
install/lib:/usr/local/lib:/opt/ros/jazzy/lib"
/home/youquan/local/bin/mc_mujoco -s -f rolling-contact-report/config/mc_rtc-ranger-mini-v3-keyboard.yaml
```

Pseudo-terminal smoke verified



Startup reported active keyboard capture; the active validation segment is retained at `/tmp/ranger-keyboard-validated_from_90_to_145.bin`. A closed-loop W trace produced a 0.300 m/s chassis reference, 0.2875246 m/s measured chassis velocity, 2.3003195 rad/s measured wheel rate against a 2.4 rad/s wheel reference, bounded 0.267145 m final position error, and no solver failures. X returned command, wheel, and chassis velocities to zero and restored the terminal cleanly. The complete test and run guide is [rolling-contact-report/README.md](#).

Known limitations

Deliberate model boundaries

- Rigid wheels with ideal/contact-mode behavior; no tire deformation, rolling resistance, or complementarity.
- A four-ray polyhedral friction cone is applied at each of two finite line-contact endpoints.
- Terrain is a supplied plane or constant-slope ramp; online terrain reconstruction is outside this scope.
- Provided examples cover differential drive and four independently steering/driving wheels.
- The planar four-steering specialization selects two independent lateral hard rows to avoid rank duplication.
- Slip and recovery thresholds require revalidation when wheel materials, terrain, or control period change.
- Timing is hardware/toolchain-specific evidence, not a portable worst-case execution-time guarantee.

EVIDENCE INDEX

Retained artifacts

The detailed report and compact JSON files retain the underlying measurements, thresholds, environment, hashes, and per-scenario results summarized here.

Complete evidence report

Human-readable implementation, validation, and reproduction narrative.

Machine-readable checkpoint

Structured revisions, gates, environment, evidence hashes, and limitations.

Physical MuJoCo summary

All 50 backend/scenario records and aggregate physical results.

10,000-cycle timing summary

Median, p95, p99, maximum, RTF, memory, and deadline results.

CPU linkage proof

Audited binaries, resolved libraries, and empty GPU runtime lists.

Validation visualization

Reproducible timing, tracking-error, and slip plots.

Completed roadmap

All stage gates, requirements, ownership, and acceptance criteria.

Architecture record

Backend integration, ownership, lifetime, and CPU-only decisions.

Run guide

CPU tests, differential/Ranger MuJoCo commands, and keyboard controls.

Physical suite SHA-256: d323fc4d1dc179d5be9786e4f2915a5571559abcdfdbf04b672cf3f10eb4ab2c

Timing suite SHA-256: 88c607a2db9a20e87770e2fe0ef2fe6f8246113d810ee5fd8af25fb568dcf818

CPU linkage SHA-256: 30bf3dae6f9e08a39d062b95d9ab1ca16619d2c816c087ea1205b9af295fa7af